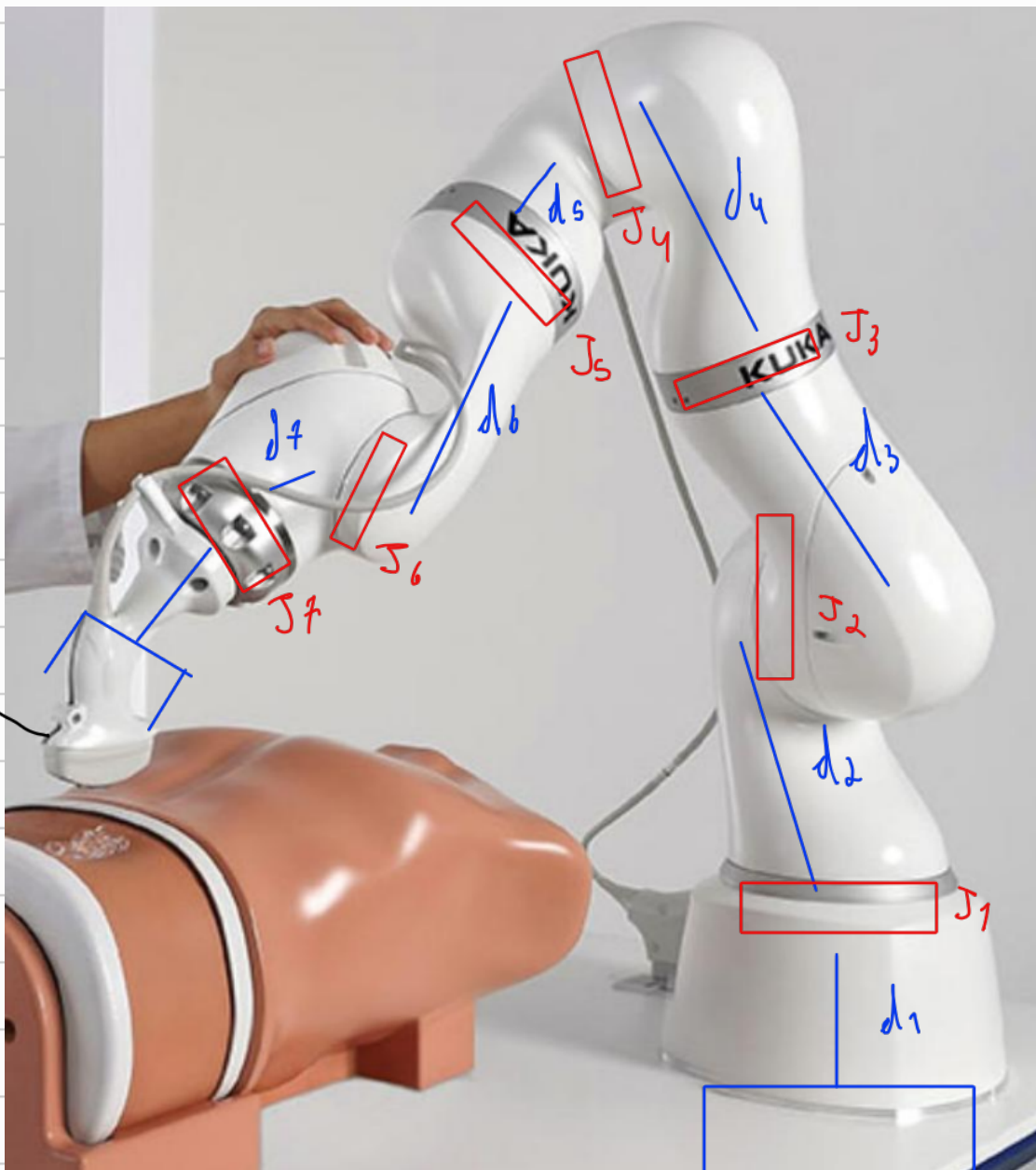


Kinematics: (Due on May 17th)

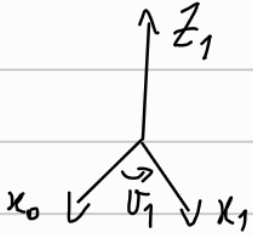
1. Analyze the robot's drawing and build its kinematic model according to the Denavit-Hartenberg convention. To do so, draw your own joints diagram with the link reference frames and the corresponding table of parameters.
2. Using the provided Toolbox, create a Simulink model for the direct kinematics of the robot. Validate your model by placing the robot in configurations for which the end-effector's position and orientation can be easily determined.
3. Following a kinematic decoupling approach present one closed-form solution for the inverse kinematics of the robot and identify the singularity conditions. Create the Simulink model and validate your model against the direct kinematics model implemented in Point 2. (NOTE: the manipulator is redundant)
4. Create a Simulink model for the geometric Jacobian of the robot. Validate the result through numerical differentiation of the direct kinematics in Point 2. Show the effect of the kinematic singularities of the robot on the rank of the Jacobian matrix.
5. Using the direct kinematics and the geometric Jacobian models, determined in Points 2 and 4, implement a Simulink model for the Closed Loop Inverse Kinematics of the robot, stabilizing the null space. Validate your model against the results of Point 3.



Joint i	d	θ	a	α	Offset
1	0	θ_1	0	$\pi/2$	0
2	0	θ_2	0	$-\pi/2$	0
3	d_3	θ_3	0	$-\pi/2$	0
4	0	θ_4	0	$\pi/2$	0
5	d_5	θ_5	0	$\pi/2$	0
6	0	θ_6	0	$-\pi/2$	0
7	d_7	θ_7	0	0	0

$\downarrow$
 angle
 They are in the same place

$$\pi/2 = 90^\circ$$



$\hookrightarrow$ rotation around the z axis

$d \rightarrow$ translation

